Lab 8: Manage and Analyse Data in a Cloud

Cloud computing brings together various technologies for users to manage, process, and analyse different types of data sets in the cloud. It supports all stages of data analysis, including development, deployment, monitoring and sharing, accelerating the production and innovation process.

This lab session will let students explore 1) how to develop a simple web application using Amazon Relational Database Service (RDS) and Amazon Elastic Compute Cloud (EC2) and 2) how to programme and run machine learning using Python in Google Collaboratory (Colab) environment for data analytics.

Topics covered

This lab will cover two parts:

- Part 1, to launch an **Amazon Relational Database Service (RDS)** and an **Elastic Compute Cloud (EC2)** to build a simple web server application for student attendance records.
- Part 2 (optional), to use Google **Collaboratory (Colab)** Service to develop a data analytics programme using Python.

Students need to complete **Part 1** to build an application using a VM and a database service, while Part 2 is optional for programming AI in the Cloud. High-level system architectures for Part 1 and Part 2 are shown below.

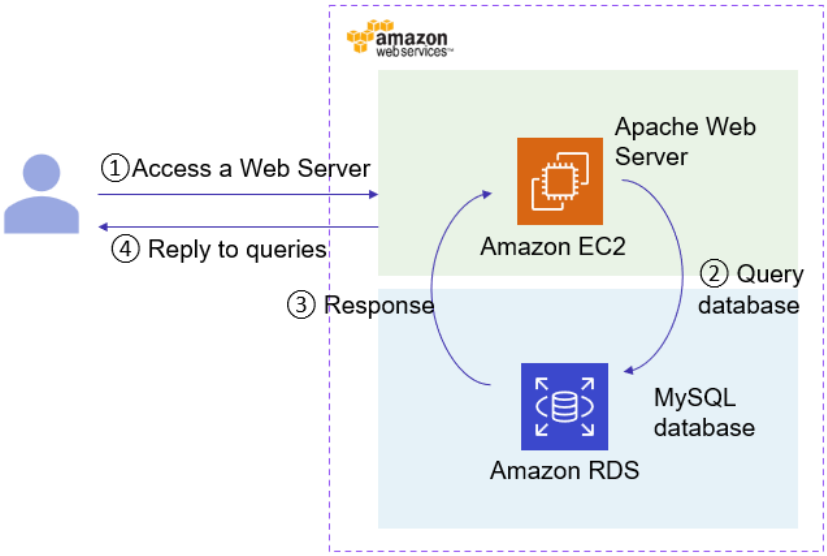


Figure 1: Part 1 - Using Amazon RDS for a Simple Web Application

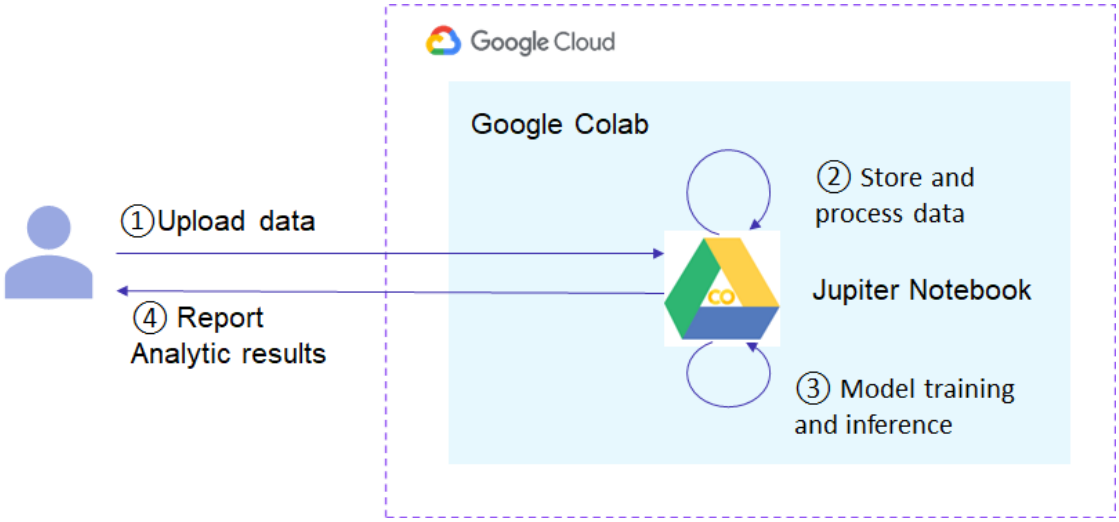


Figure 2: Part-2 - Using Google Colab to develop a Machine Learning program

Learning Objectives

- Launching an Amazon EC2 Web Server instance.
- Launching an Amazon Relational Database Service (RDS) instance
- Developing a simple web application for recording attendance
- Sign in to the Google Colaboratory (Colab) web portal
- Creating a Jupyter Notebook
- Running data analytics with Python

Duration

This lab takes approximately **110 minutes** to complete.

Part 1: Build a Web Application for Student Attendance Recording with Amazon RDS

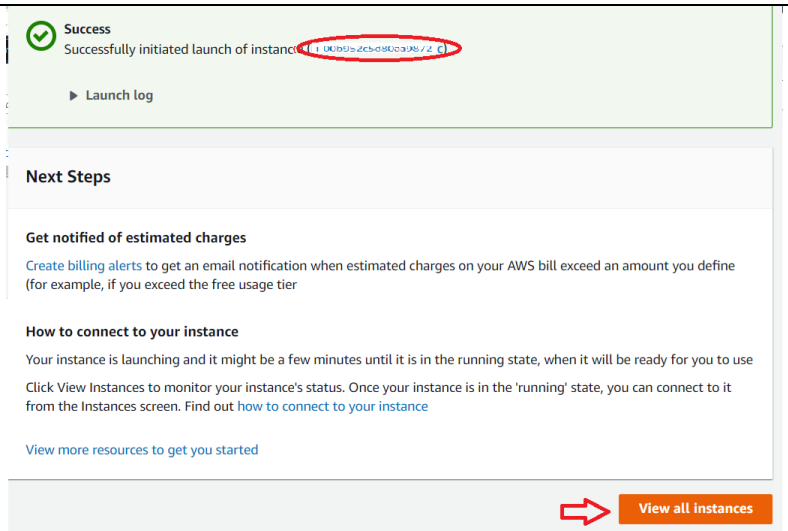
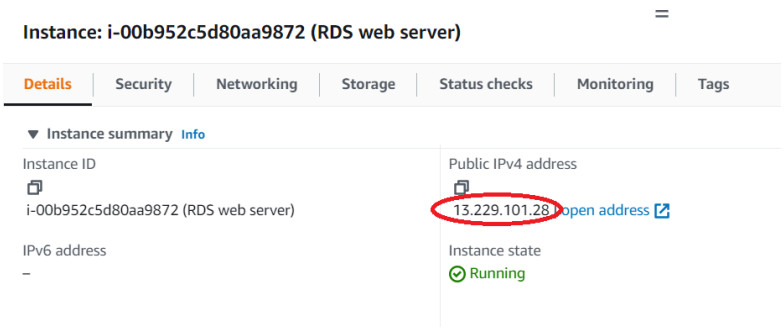
In Part 1, students will practise creating a web service with which users can modify and query Amazon Relational Database (RDS) services to record the attendance of a class. The web server will run on an Amazon EC2 instance and use Amazon RDS database service as a MySQL database.

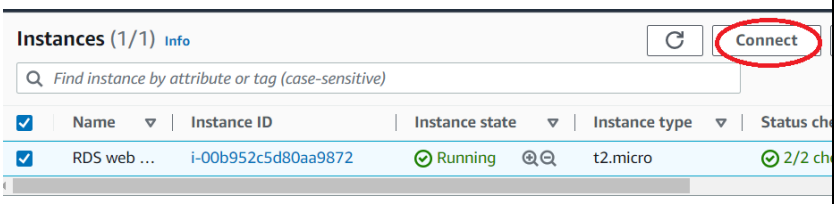
Task 1: Create a Web Server

Go to **AWS Management Console**, enter EC2 in the search box of Services, select **EC2,** and choose **Launch instances**

SN	Steps	Instructions
1)	Name and Tag: Name the EC2 instance as **"RDS Web Server"**	**Name and tags** Info Name RDS Web Server
2)	Choose the default **AMI**	**Application and OS Images (Amazon Machine Image)** Info An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below
3)	Choose the default instance type **t2.micro**	**Instance type** Info Instance type t2.micro Free tier eligible Family: t2 1 vCPU 1 GiB Memory On-Demand Linux pricing: 0.0146 USD per Hour On-Demand Windows pricing: 0.0192 USD per Hour Compare instance types
4)	Choose the default key pair: **vockey**	**Key pair (login)** Info You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance. Key pair name - *required* vockey Create new key pair
5)	Network Settings	**Network settings** Info Network Info vpc-0a030916b83943389 Subnet Info No preference (Default subnet in any availability zone) Auto-assign public IP Info Enable **Firewall (security groups)** Info A security group is a set of firewall rules that control the traffic for your instance. Add rules to allo instance. Create security group Select existing security group We'll create a new security group called '**launch-wizard-7**' with the following rules: Allow SSH traffic from Helps you connect to your instance Anywhere 0.0.0.0/0 Allow HTTPs traffic from the internet To set up an endpoint, for example when creating a web server Allow HTTP traffic from the internet To set up an endpoint, for example when creating a web server

6)	Expand advanced details to configure the user data	▼ **Advanced details** Info User data Info Copy the following scripts into User data. #!/bin/bash # Update package and start Apache web server sudo yum update -y sudo yum install httpd -y # Create a simple index.html file echo '<html><h1>Hello From Web Server 1!</h1></html>' > a.html sudo cp a.html /var/www/html/index.html rm a.html # Set appropriate permissions sudo chown apache:apache /var/www/html/index.html sudo chmod 644 /var/www/html/index.html #install php sudo yum install php-cli php-pdo php-fpm php-json php-mysqlnd -y #install mysql client sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm sudo dnf install mysql80-community-release-el9-1.noarch.rpm -y sudo rpm --import https://repo.mysql.com/RPM-GPG-KEY-mysql-2023 sudo dnf install mysql-community-client -y #start the web server sudo systemctl start httpd sudo systemctl enable httpd # Display server status sudo systemctl status httpds

		The *user data* will perform automated installation and configuration tasks after the instance starts. The script will install an Apache web server and create a simple web page.
7)	Launch instance	
8)	View all instances	
9)	Get the public IP address	Once the instance has been successfully initialised, get its public IP address.  Open a web browser and check the URL http://<public_ip_address>
10)	Connect to the EC2 instance via SSH	Select RDS web server in the right pane and choose Connect. In the EC2 Instance Connect tab, choose **Connect**.

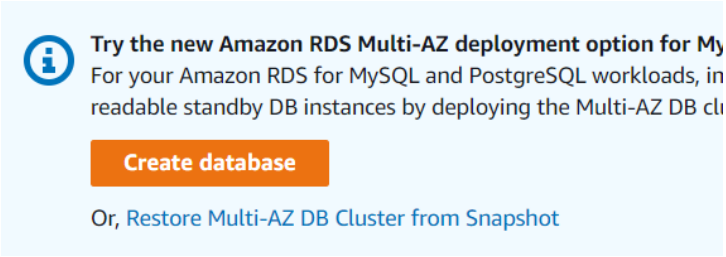
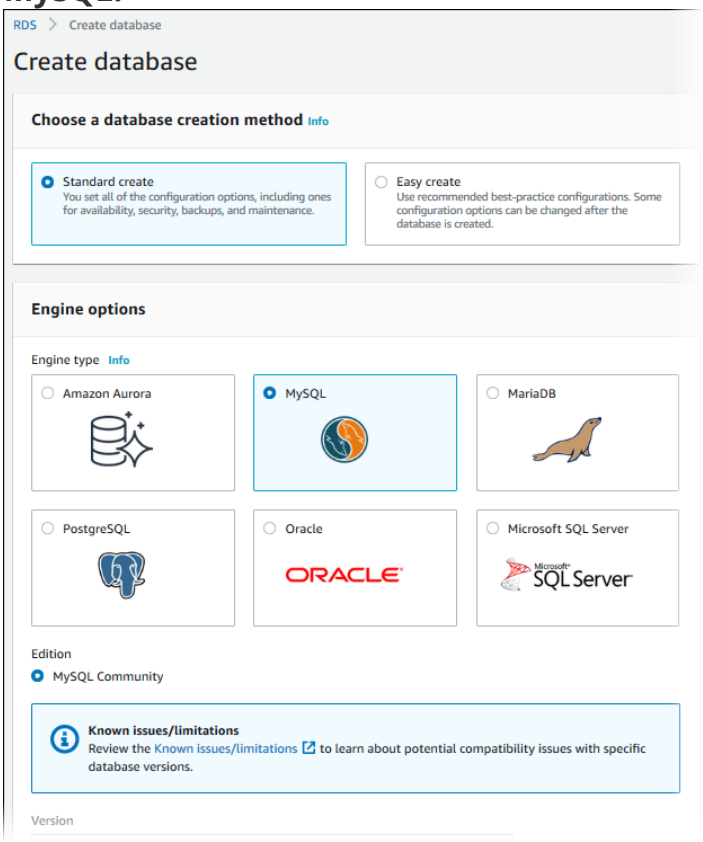
		

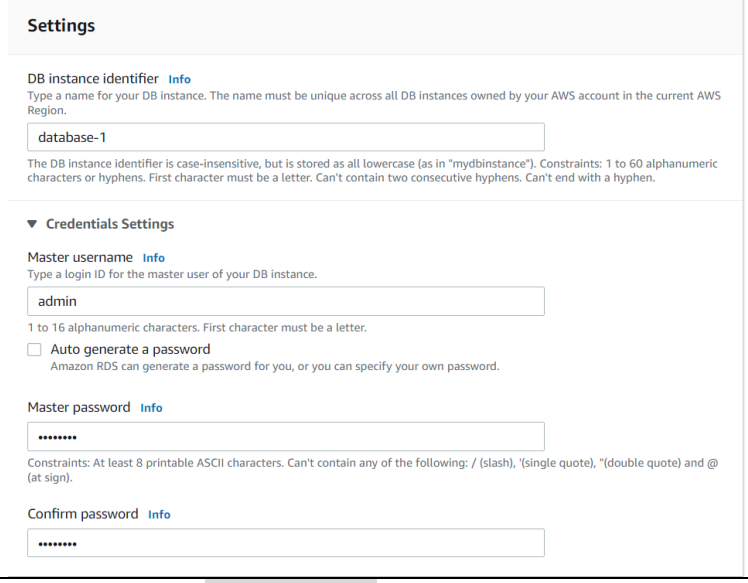
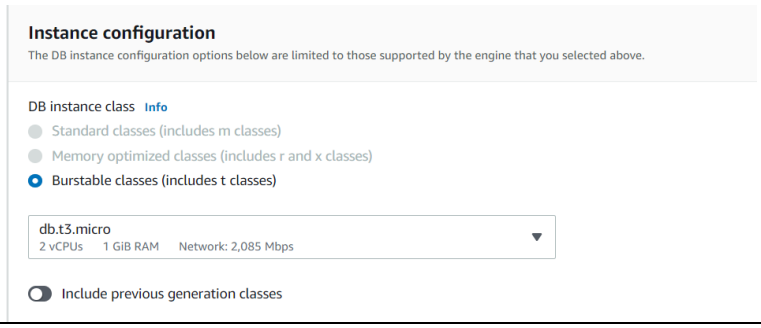
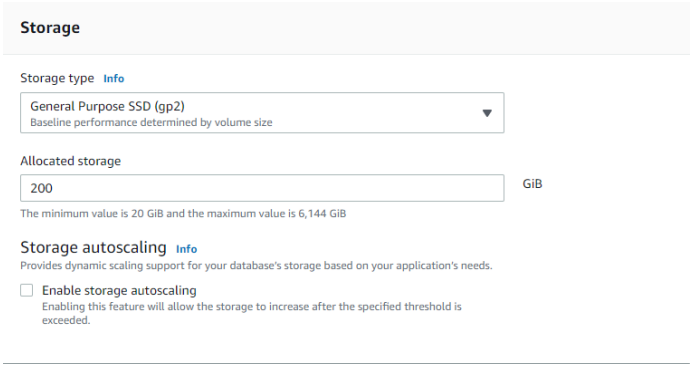
10)	Check if PHP and MySQL is properly installed on the EC2 instance	1. In the EC2 Instance command line window, check if php and mysql have been installed properly.
		
		`$php --version`
		
		It should show the version of the installed php.
		
		`$mysql --version`
		
		It should show the version of the installed MySQL client.
		
		If not, please run the following scripts one by one.
		
		#!/bin/bash
		# Update package and start Apache web server
		sudo yum update -y
		sudo yum install httpd -y
		
		# Create a simple index.html file
		echo '<html><h1>Hello From Web Server 1!</h1></html>' > a.html
		sudo cp a.html /var/www/html/index.html
		rm a.html
		
		# Set appropriate permissions
		sudo chown apache:apache /var/www/html/index.html
		sudo chmod 644 /var/www/html/index.html
		
		#install php
		sudo yum install php-cli php-pdo php-fpm php-json php-mysqlnd -y
		
		#install mysql client
		sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-1.noarch.rpm
		sudo dnf install mysql80-community-release-el9-1.noarch.rpm -y
		sudo rpm --import https://repo.mysql.com/RPM-GPG-KEY-mysql-2023
		sudo dnf install mysql-community-client -y
		
		#start the web server
		sudo systemctl start httpd
		sudo systemctl enable httpd
		
		# Display server status
		sudo systemctl status httpd

		2. Open a web browser and check the url http://<public_ip_address>

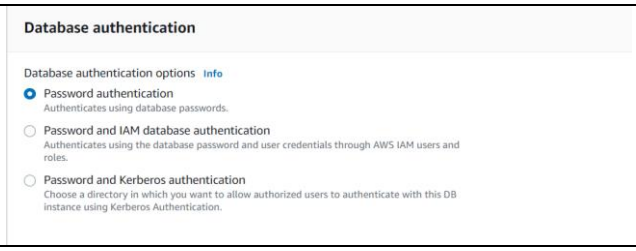
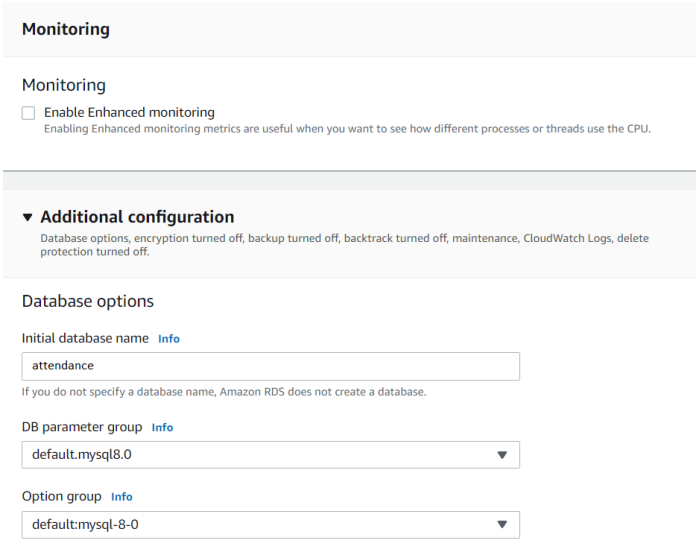
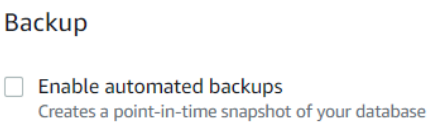
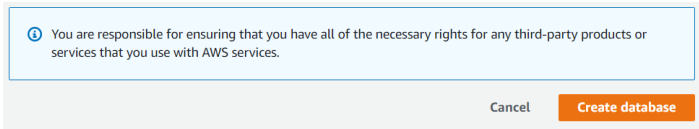
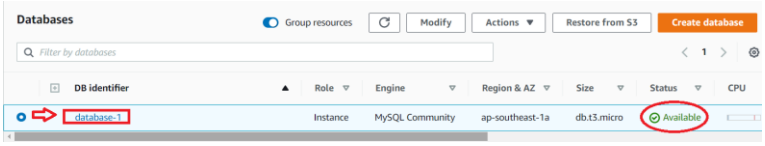
Task 2: Create an RDS database

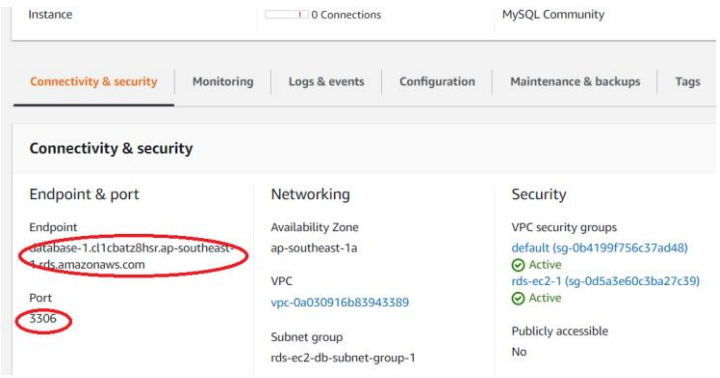
Go to the search box of **Services** and enter **RDS** and select **RDS**.

SN	Steps	Instructions
1)	Create **RDS database**	Choose **Create database** 
2)	Choose MySQL engine	In the RDS page, choose **Standard create. Select MySQL.** 

2)	Select Templates: **Free tier**	
3)	Instance name: **database-1** Username: **admin** Set your Password: **testtest**	
4)	Instance Configuration	Use the default db.t3.micro 
5)	Storage	Uncheck Enable storage autoscaling. 
6)	Connectivity	Select Connect to an EC2 compute resource

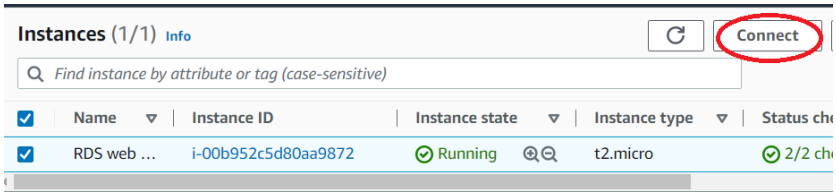
		And choose EC2instance ID for the RDS Web Server instance we just created. **Connectivity** Info Compute resource Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database. ○ Don't connect to an EC2 compute resource — Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later. ◉ Connect to an EC2 compute resource — Set up a connection to an EC2 compute resource for this database. EC2 Instance Info Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database. i-00b952c5d80aa9872 RDS web server Select Automatic setup for DB subnet group Virtual private cloud (VPC) Info Choose the VPC. The VPC defines the virtual networking environment for this DB instance. Default VPC (vpc-010887e9b343a90c6) Only VPCs with a corresponding DB subnet group are listed. After a database is created, you can't change its VPC. DB subnet group Info Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected. ○ Choose existing — Choose existing DB subnet group ◉ Automatic setup — RDS creates a new subnet group for you or reuses an existing subnet group Choose Create new to create a new VPC security group and name it as MyDBSecurityGroup. VPC security group (firewall) Info Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic. ○ Choose existing — Choose existing VPC security groups ◉ Create new — Create new VPC security group New VPC security group name MyDBSecurityGroup Amazon RDS will add a new VPC security group *rds-ec2-1* to allow connectivity with your compute resource. Availability Zone Info Certificate authority - *optional* Info Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision. rds-ca-2019 (default) If you don't select a certificate authority, RDS chooses one for you.
7)	Authentication	Use the default Password authentication

		
8)	Additional Configuration: Set initial database name: **attendance**	After the monitoring section, there is another section "Additional Configuration". Expand the section to set the initial database name as attendance.  Disable Enable Automated backups 
10)	Create the database	Choose **Create database** 
11)	Check the status	It may take a few minutes for the RDS to be available. When the new DB instance is Available, click database-1 to see the details. 

12)	Note the Endpoint	In the Connectivity & security tab of database1, view the Endpoint and Port of the DB instance.  **Add the screenshot into Result 1 in the lab report**

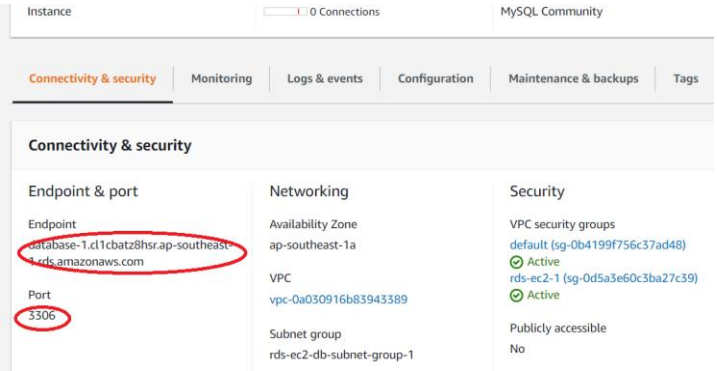
Task 3: Configure the web server and query the RDS instance

Go to search box in **Services**, select **EC2** and go to **instances** in the left pane.

SN	Steps	Instructions
1)	Connect to the EC2 instance	Select RDS web server in the right pane and choose Connect. In the EC2 Instance Connect tab, choose **Connect**. 

2)	Set permissions for the ec2-user to access Apache server document	1. Change the group ownership of the /var/www directory and its contents to the apache group.
		`$sudo chown -R ec2-user:apache /var/www`
		
		2. Change the directory permissions of /var/www and its subdirectories to add group write permissions and set the group ID on subdirectories created in the future.
		
		`$sudo chmod 2775 /var/www`
		`$find /var/www -type d -exec sudo chmod 2775 {} \;`
		
		3. Recursively change the permissions for files in the /var/www directory and its subdirectories to add

		group write permissions. `$find /var/www -type f -exec sudo chmod 0664 {} \;`

3)	Connect EC2 to RDS database	1. Log in to MySQL with the RDS endpoint and user name
		
		Go to the Connectivity & security tab of database1, copy the endpoint e.g., *database-1.cl1cbatz8hsr.ap-southeast-1.rds.amazonaws.com. Replace* *<endpoint>* using database1 endpoint.
		
		`$mysql -h <endpoint> -u admin -p`
		
		 
		
		2. Enter the RDS password (i.e., *testtest*) and it should show the MySQL interactive interface
		
		`MySQL [(none)]>>_`
		
		3. Check the database;
		
		`MySQL [(none)]>Show databases;`
		
		`+--------------------+`
		`| Database |`
		`+--------------------+`
		`| attendance |`
		`| information_schema |`
		`| mysql |`
		`| performance_schema |`
		`| sys |`
		`+--------------------+`
		`5 rows in set (0.002 sec)`
		
		It should include the attendance database that we just created in database1 RDS service.
		
		Add the screenshot to Result 2 in the lab report

		4. Exit from MySQL; `MySQL [(none)]>>exit`

Task 4: Program the web server page with PHP

Follow the steps:

SN	Steps	Instructions

1)	Create a web server config file	1. Under the EC2 instance Connect shell, create a file named `dbinfo.inc` using `vim`.
		
		`$vim dbinfo.inc`
		
		Copy the following content to `dbinfo.inc`
		
		`<?php`
		
		`define('DB_SERVER', '<db_instance_endpoint>');`
		`define('DB_USERNAME', 'admin');`
		`define('DB_PASSWORD', 'testtest');`
		`define('DB_DATABASE', 'attendance');`
		
		`?>`
		
		Replace `<db_instance_endpoint>` with the database1 RDS endpoint (e.g., database-1.cl1cbatz8hsr.ap-southeast-1.rds.amazonaws.com). The RDS endpoint can be copied from the RDS instance.
		
		2. Create a folder under /var/www and copy the file to the folder
		
		`$mkdir /var/www/inc`
		`$cp dbinfo.inc /var/www/inc`

2)	Create a PHP file for the	1. Create a file named `attendance.php` `$vim attendance.php`

<table>
<tr><td></td><td>web page</td><td>Download <code>attendance.php</code> from x-site or from github at https://github.com/csd3156/csd3156-lab8-2023/tree/main/Part%201 and copy the content into <code>attendance.php</code>

2. Copy <code>attendance.php</code> file to <code>/var/www/html</code>

<code>$cp attendance.php /var/www/html</code></td></tr>
<tr><td>3)</td><td>Go to the webpage and add the attendance</td><td>1. Go to the web browser and paste the url below. The <public_ip> is the ip address or the end point of the EC2 instance.

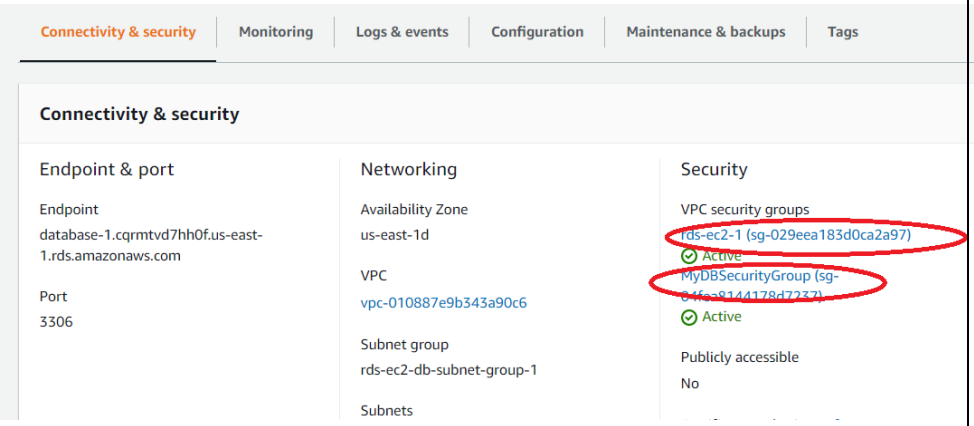
http://<public_ip>/attendance.php

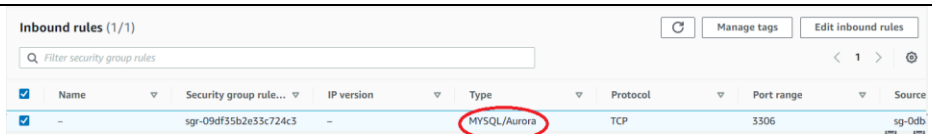
2. Add the attendance with STUDENTID and NAME and click Add

<table>
<tr><td>STUDENTID</td><td>NAME</td></tr>
<tr><td>123</td><td>Julia</td></tr>
</table>

Add the screenshot to Result 3 the lab report

Troubleshooting 1: If the webpage http://<public_ip>/attendance.php does not work, check the Connectivity & Security of the database1 if the VPC security groups and MyDBSecurityGroup allow MySQL/Aurora traffic in the inbound rules.


</td></tr>
</table>



Trouble shooting 2: If the webpage still does not work, connect to RDS Web Server EC2 instance and check if PHP is properly installed.

```
$php --version
```

If PHP is properly installed, restart& enable the Apache server.

```
$sudo systemctl restart httpd
$sudo systemctl enable httpd
```

Check if http://<public_ip> shows "Hello From Web Server!"

Trouble shooting 3: if http://<public_ip> works but http://<public_ip>/attendance.php does not, connect to RDS Web Server EC2 instance and check if <db_instance_endpoint> was replaced with the database1 RDS endpoint (e.g., database-1.cl1cbatz8hsr.ap-southeast-1.rds.amazonaws.com).

```
$cat dbinfo.inc
```

```
<?php

define('DB_SERVER', '<db_instance_endpoint>');
define('DB_USERNAME', 'admin');
define('DB_PASSWORD', 'testtest');
define('DB_DATABASE', 'attendance');

?>
```

Check if the application file attendance.php has been copied to folder `/var/www/html`

```
$ls /var/www/html/attendance.php
```

Check if the database configuration file `dbinfo.inc` has been copied to folder `/var/www/inc`

```
$ls /var/www/inc/var/www/inc
```

Task 5: Stop the services and clear the resources

SN	Steps	Instructions
1)	Stop RDS	In search box of **Services,** enter **RDS** to go into RDS management console. Select **Databases** in left navigation pane, select database-1. In Action menu, select **Delete**. Uncheck Create final snapshot? and check I acknowledge that upon instance deletion, automated backups, including system snapshots and point-in-time recovery, will no longer be available. Type Delete me to delete the database.
2)	Stop EC2	In search box of **Services,** key in EC2 to go into EC2 management console. Select **Instances** in left navigation pane, select RDS Web Server. In Instance state menu, select **Terminate instance**.

Add one or two sentences to summarize what you have learned from Lab Part 1 into the lab report.

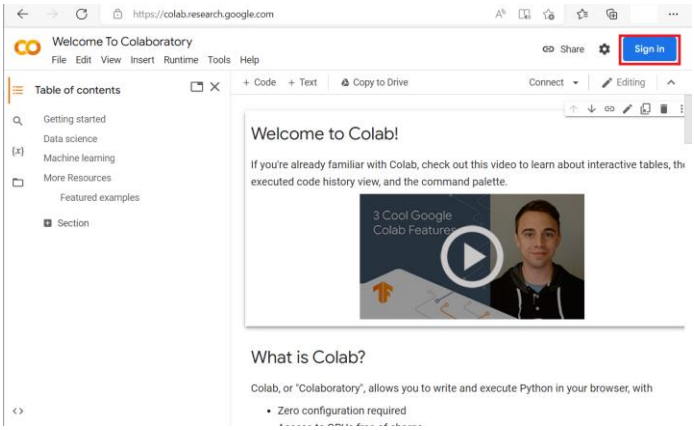
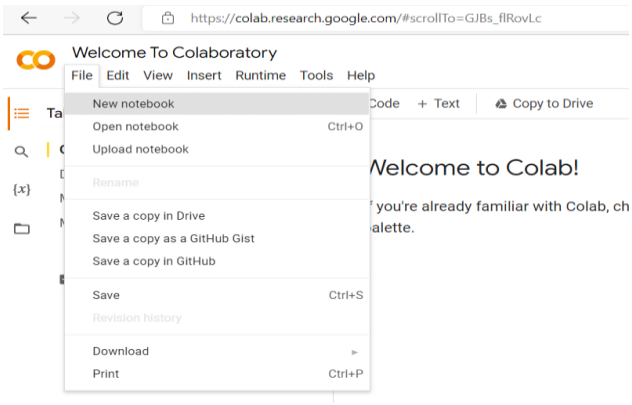
Part 2: Use Google Colab for Data Analytics

Google Colaboratory, or Google Colab for short, is a software-as-a-service platform for cloud analytics. It is a free resource that allows users to run and execute Python code from a web browser without installing any software. It is lightweight and suitable for data analytics and machine learning especially. It can also connect with other Google Cloud resources e.g., TensorFlow, GPU, Google BigQuery, etc. to power data analytics.

In Part 2, students can run a Python code as an example of machine learning on Google Colab. You can run through the steps of the Python code to train a machine learning model for diabetes estimation.

Task 1: Login into Google Colab (Colaborary) and create a Jupiter Notebook

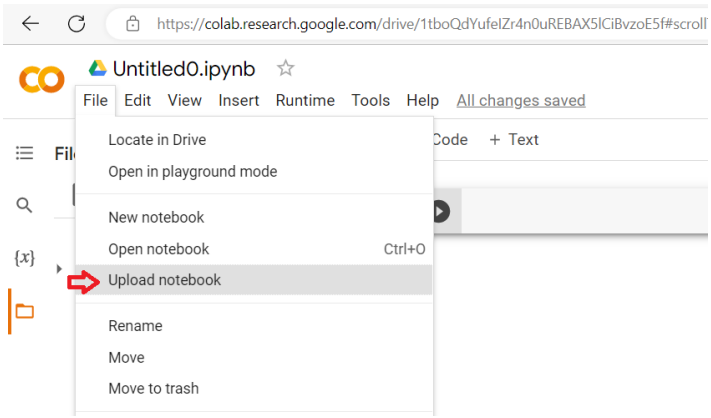
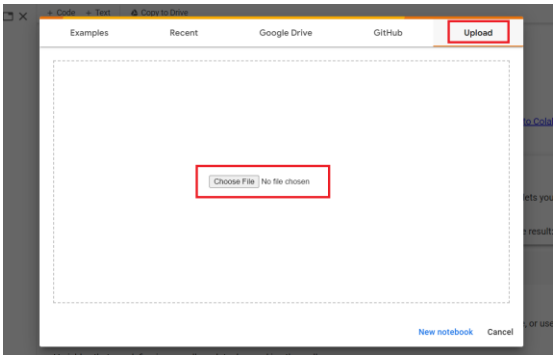
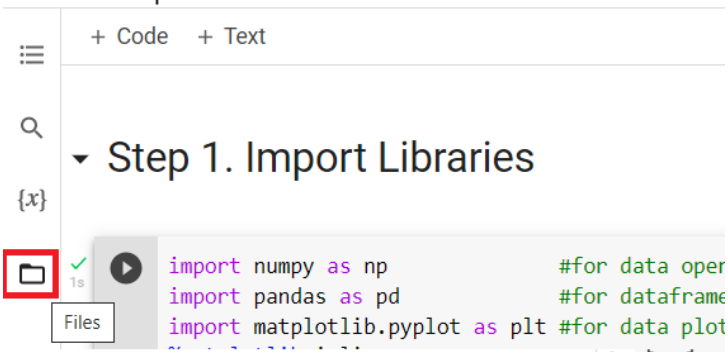
Go to https://colab.research.google.com and sign in or log into Google Colab

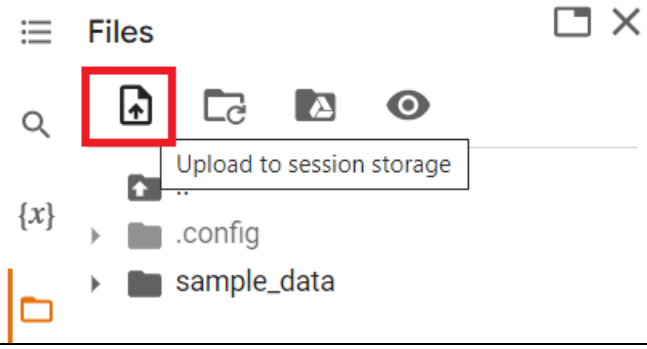
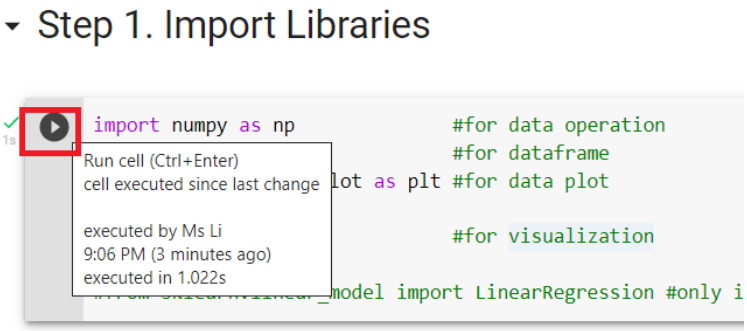
SN	Steps	Instruction
1)	Sign in Google Colab	Click Sign in at the right top. You may need to create an account or use your GCP account to log in. 
2)	Create a new notebook	Create a new notebook by choose File->New notebook. 

Task 2: Run through the machine learning codes for diabetes estimation

You can follow the steps:

SN	Steps	Instructions

1)	Upload the **diabetes-analysis.ipynb**	Download **diabetes-analysis.ipynb** from x-site or from github at https://github.com/csd3156/csd3156-lab8-2023/tree/main/Part%202 Upload **diabetes-analysis.ipynb**  
1)	Load the data file **diabetes-data.csv**	Download the data file, **diabetes-data.csv**, from github at https://github.com/csd3156/csd3156-lab8-2023/tree/main/Part%202 Upload the data file by choosing the folder icon in the left pane. 

		Upload **diabetes-data.csv**. 
3)	Run through the codes	Run the code by clicking the playback icon from Step 1 to Step 4. 

Task 3: Program the Python code

Modify the code to improve the data analytics

1. Add a line of code in Step 2 to remove the duplicates
2. Rerun Steps 3 and 4 and compare the previous results

Add one or two sentences to summarize what you have learned from the lab Part 2 into the lab report.

Lab complete

Congratulations! You have completed the lab.

Please submit the lab report to the x-site by the end of 24 Mar 2026 (Tuesday).

Thank you!